

DIGITAL FORENSICS & INCIDENT RESPONSE

CONFIDENTIAL FORENSIC ANALYSIS

Digital Forensic Analysis of base-wkstn-01-c-drive.E01

Host base-wkstn-01.shieldbase.lan - E01 disk image acquired
2021-09-16

CLIENT	shieldbase.lan (host base-wkstn-01)
CASE ID	CASE-001 / EVID-001
PREPARED BY	DFIR Orchestrator (autonomous SIFT)
REPORT DATE	2026-06-04 UTC
CLASSIFICATION	CONFIDENTIAL — RESTRICTED DISTRIBUTION

■ **CONFIDENTIAL**

Report generated 2026-06-04

01 Executive Summary

Digital forensic analysis of the EVID-001 E01 disk image of host base-wkstn-01.shieldbase.lan (acquired 2021-09-16 03:10:21 UTC by FTK Imager v4.2.0.13) identified **multiple confirmed malicious software detections and one confirmed instance of offensive PowerShell script execution** on the host between 2018 and 2019.

Confirmed observations on EVID-001:

1. **86 McAfee VirusScan Enterprise (VSE) On-Access Scanner trojan detections** in the on-host AV log between 2019-05-08 and 2019-11-06. All detected files were deleted by McAfee at detection time.
2. **5 residual binaries remain on disk** in C:\ProgramData\perfmon-k\ co-located with the directory in which McAfee detected the Artemis Trojan family in May 2019. These residual files were not flagged by McAfee at detection time.
3. **A staging directory** at C:\ProgramData\staging\ contains the 7-Zip command-line archiver (7za.exe), two installer binaries named to mimic Windows advapi2 (install_msadvapi2_32.exe, install_msadvapi2_64.exe) under a subdirectory named install_wormhole, an unknown 138 MB installer (Lariat-9.4.1-install.exe), Puppet mcollective_agents zip archives, and three certificate files whose subject names parody VeriSign.
4. **One confirmed offensive PowerShell shellcode-injection script** was executed on 2018-09-06 17:13:58 - 17:14:05 UTC by the domain user S-1-5-21-3445421715-2530590580-3149308974-1193 (account not present on this hosts local profile list). The script block content recorded by PowerShell EID 4104 contains the func_get_proc_address and var_unsafe_native_methods reflective injection scaffold characteristic of PowerSploits Invoke-Shellcode / Invoke-ReflectivePEInjection.
5. **A privileged-group modification** occurred on 2020-02-17 08:00:37 UTC: domain SID S-1-5-21-3445421715-2530590580-3149308974-1212 was added to BUILTIN\Administrators by the host machine account (system).
The terms Compromised host, Pass-the-hash, Exfiltration confirmed, and Attacker are not used in this report unless the evidence threshold is met.

02 Evidence Examined

EVIDENCE ID	SOURCE PATH	TYPE	SIZE (BYTES)	MD5	SHA1	VERIFICATION
EVID-001	/cases/Test1/	EWf E01 (FTK)	33,833,353,216	fd463e4eb6b574	c4e37e7fe2d95d	ewfverify

EVIDENCE ID	SOURCE PATH	TYPE	SIZE (BYTES)	MD5	SHA1	VERIFICATION
	base-wkstn-01 -c- drive.E01	Imager v4.2.0.13)		4c84b5a d138e7fe c7f	b50d3c5 a8687a5 cf6aad10 1503	SUCCESS (stored vs computed MD5 match)

Acquisition metadata recorded in the E01: Case number 001, Description base-wkstn-01 c-drive, Acquisition date 2021-09-16 03:10:21, OS recorded Win 201x.

Mount record (./analysis/mounts.csv):

EVIDENCE ID	EWFF MOUNT	FILESYSTEM MOUNT	OPTIONS	STATUS
EVID-001	/mnt/ ewf_wkstn01	/mnt/wkstn01	ro,noatime,fuseblk (verified read-only)	Active read-only during analysis

NTFS metadata: Volume Serial Number 82424B85424B7CC3; Cluster Size 4096; Total clusters 0-8260094 (~31 GiB).

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Tools and Versions

TOOL	VERSION / PATH	USE
ewfinfo / ewfverify / ewfmount	libewf 20140816	Evidence verification and read-only mount
mmls / fsstat / icat	The Sleuth Kit	Partition table, filesystem metadata, file extraction
EvtxECmd	/opt/zimmermantools/ EvtxeCmd/EvtxECmd.dll (dotnet 6.0.36)	Event log parsing
AppCompatCacheParser	/opt/zimmermantools/ AppCompatCacheParser.dll	Shimcache parser
AmcacheParser	/opt/zimmermantools/ AmcacheParser.dll	Amcache.hve parser
RECmd (Kroll_Batch.reb)	/opt/zimmermantools/ RECmd/RECmd.dll	Registry batch parser
MFTECmd	/opt/zimmermantools/ MFTECmd.dll	MFT parser
Python 3.12	/usr/bin/python3	

TOOL	VERSION / PATH	USE
		EVTX CSV post-filtering and IOC register construction

Tools not available on this SIFT instance: PECmd.dll, SrumECmd.dll, /usr/local/bin/yara. Prefetch was not parsed; SRUM SRUDB.dat was extracted but not parsed. YARA scanning was not executed; on-disk IOC matching was performed via sha256sum and md5sum directly.

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Timeline of Confirmed Activity (UTC)

UTC TIME	ACTIVITY	SOURCE ARTIFACT
2018-05-04 18:14:46	Host base-wkstn-01 initial WDAGUtilityAccount password set	SAM hive
2018-05-07 19:24:57	Local Administrator/Guest/DefaultAccount/WDAGUtility built on host	SAM hive
2018-05-11 22:08:34	Local profile created for mhill (domain SID -1118)	SOFTWARE\ProfileList
2018-07-06 01:50:31	Local profile created for rsydow-a (domain SID -1183)	SOFTWARE\ProfileList
2018-08-15 15:32:50	Local profile created for cbarton-a (domain SID -1185)	SOFTWARE\ProfileList
2018-09-06 17:13:58 - 17:14:05	Offensive PowerShell shellcode-injection script executed (3 distinct EID 4104 records); user SID -1193	Microsoft-Windows-PowerShell/Operational.evtx, EID 4104
2019-05-08 15:14:04	McAfee VSE deletes C:\ProgramData\perfmon-k\perfmon-kconfigure.exe (Artemis!1977C126E8E7)	McAfee OnAccessScanLog.txt
2019-05-17 22:29:13	McAfee VSE deletes C:\ProgramData\perfmon-k\perfmon-k.exe (Artemis!50CCE7D590CA)	McAfee OnAccessScanLog.txt
2019-10-22 20:55:41 to 2019-10-24 15:55:59	31 McAfee VSE deletions of chrome_updater.exe (GenericRXIT-PW!	McAfee OnAccessScanLog.txt

UTC TIME	ACTIVITY	SOURCE ARTIFACT
	264609C34194) from C:\Users\mhill\AppData\Local\Temp\{GUID}; user shieldbase\mhill; parent svchost.exe	
2019-11-04 19:55:52 to 2019-11-06 17:01:44	38 McAfee VSE deletions of chrome_updater.exe (GenericRXIT-PW! D870AC37E98C)	McAfee OnAccessScanLog.txt
2019-11-06 22:01:05 to 2019-11-06 22:01:54	6 McAfee VSE deletions of chrome_updater.exe (GenericRXIT-PW! 4D17F453B454)	McAfee OnAccessScanLog.txt
2019-12-22 11:15:00 (local)	Sysmon Update scheduled task registered by shieldbase\rsydow-a, running as SYSTEM (Sysmon EDR deployed)	Sysmon Update task XML
2020-02-17 08:00:37	Domain user SID -1212 added to BUILTIN\Administrators by host machine account	Security EID 4732
2020-09-21 01:01:35	Defender real-time protection disabled (EID 5001) - consistent with McAfee being the active AV	Microsoft-Windows-Windows Defender/Operational.evtx
2021-02-03 21:51:36	Last logged-on SAM user recorded as shieldbase\mhill (LogonUI key)	SOFTWARE hive
2021-09-16 03:10:21	E01 acquisition completed by FTK Imager	ewfinfo metadata

User SID resolution:

SID	PROFILE PATH	MAPS TO
S-1-5-21-3225607321-4158839294-2596801906-500	C:\Users\Administrator.BASE-WKSTN-01	Original local Administrator
S-1-5-21-3445421715-2530590580-3149308974-500	C:\Users\administrator.shieldbase	Domain Administrator
	C:\Users\mhill	Domain user mhill

SID	PROFILE PATH	MAPS TO
S-1-5-21-3445421715-253 0590580-3149308974-111 8		
S-1-5-21-3445421715-253 0590580-3149308974-118 3	C:\Users\rsydow-a	Domain user rsydow-a
S-1-5-21-3445421715-253 0590580-3149308974-118 5	C:\Users\cbarton-a	Domain user cbarton-a
S-1-5-21-3445421715-253 0590580-3149308974-119 3	(no local profile)	Ran the 2018-09-06 PowerShell; not resolvable from this host
S-1-5-21-3445421715-253 0590580-3149308974-121 2	(no local profile)	Added to local Administrators on 2020-02-17

05 Findings

Finding F-001 - McAfee VSE detected and deleted Artemis Trojan binaries in C:\ProgramData\perfmon-k\

- **Confidence:** Confirmed
- **Claim Type:** Direct Observation
- **Evidence:** ./exports/mcafee/OnAccessScanLog.txt
- **Tool Output:** ./logs/20260604T210625Z_17741_stdout.log, ./logs/20260604T205909Z_12853_stdout.log
- **Timestamps (UTC):**
 - 2019-05-08 15:14:04 - Artemis!1977C126E8E7 (Trojan) - MD5 1977c126e8e74da4460f41653ca6a6f2 - C:\ProgramData\perfmon-k\perfmon-kconfigure.exe
 - 2019-05-17 22:29:13 - Artemis!50CCE7D590CA (Trojan) - MD5 50cce7d590ca72cfb1f3b5522ea5c27b - C:\ProgramData\perfmon-k\perfmon-k.exe
- **Process context:** Detecting process recorded as C:\Windows\System32\CompatTelRunner.exe, user NT AUTHORITY\SYSTEM.
- **AV action:** Deleted in both events.
- **Alternative explanation:** McAfee Artemis! is a cloud-lookup heuristic and can false-positive; however the two distinct MD5s, both with the family prefix Artemis! and the matching MD5 substring, the staging in a non-system directory, and the residual files left in the same directory all point away from a false positive.
- **Limitation:** McAfee log records the detection, not the executing parent.
- **What evidence would resolve uncertainty:** Sysmon or 4688 process-create events from before 2019-05 - not present (Sysmon was deployed later on 2019-12-22).

Finding F-002 - Residual perfmon-k binaries remain on disk

- **Confidence:** Probable
- **Claim Type:** Supported Inference
- **Evidence:** On-disk hashes computed by sha256sum/md5sum (./logs/20260604T205838Z_10853_stdout.log)
- **On-disk component table** (SHA-256 column is truncated to first 16 chars; full values in artifact_manifest.csv):**

FILE PATH	SIZE BYTES	MD5	SHA-256	EVIDENCE SOURCE	NOTES
C:\ProgramData\perfmon-k\perfmon-khk.dll	1307648	7cfcf2e7996ce96b44013e5fe0705820	225efff946151ede...	EVID-001	Co-located with deleted Artemis trojans
C:\ProgramData\perfmon-k\perfmon-ki.dll	275968	d567a6d0647f80ecb5a761ddd9ad367c	fa76d9059fc4fe10...	EVID-001	Co-located residual
C:\ProgramData\perfmon-k\perfmon-kr.exe	1352704	2d444c6afb6d647d246d2eb01a9e75c9	fc7f5094bc182a05...	EVID-001	Co-located residual EXE
C:\ProgramData\perfmon-k\perfmon-kvw.exe	1570816	2c96b03ff718bd2cdf4288864edeb6a0	128cdb7c58bfda11...	EVID-001	Co-located residual EXE
C:\ProgramData\perfmon-k\perfmon-kwb.dll	1348608	2ff0e7f97038863c4448bc08eb4286c8	8db5e3bd57ed77f...	EVID-001	Co-located residual DLL
C:\ProgramData\perfmon-k\install.bin	83456	a7e4adb5314851ff3899908875c0667c	0190791522d25607...	EVID-001	Installer artifact
C:\ProgramData\perfmon-k\helpfile.txt	407279	a3838f8bbf585d3c5c	beb9c5972883d924...	EVID-001	Help file

FILE PATH	SIZE BYTES	MD5	SHA-256	EVIDENCE SOURCE	NOTES
ata\perfmo		fc3b615fcc			
n-		50c9			
k\help.chm					

- **Why probable not confirmed:** These files were not flagged by McAfee at detection time. The strength of the association is positional (same directory as confirmed Artemis Trojan drops). No Sysmon event mentions perfmon-k (Sysmon installed 2019-12-22, after the May 2019 detections).
- **Alternative explanation:** A legitimate third-party performance-monitoring product could share the perfmon-k name.
- **What evidence would resolve uncertainty:** PE static analysis (Authenticode signer, imports, strings) or VirusTotal lookup of the SHA-256s.

Finding F-003 - McAfee VSE detected and deleted 84 chrome_updater.exe Trojan drops in mhill Temp directory

- **Confidence:** Confirmed
- **Claim Type:** Direct Observation
- **Evidence:** ./exports/mcafee/OnAccessScanLog.txt
- **Tool Output:** ./logs/20260604T205933Z_21809_stdout.log
- **Detected file MD5s:**
 - 264609c34194d47dbf25d18d4f83d41d → GenericRXIT-PW!264609C34194 (31 deletions, 2019-10-22 to 2019-10-24)
 - d870ac37e98c1351fd0de3ae404a180d → GenericRXIT-PW!D870AC37E98C (38 deletions, 2019-11-04 to 2019-11-06)
 - 4d17f453b454859f249990be61f36091 → GenericRXIT-PW!4D17F453B454 (6 deletions, 2019-11-06)
- **All detections share the path template:** C:\Users\mhill\AppData\Local\Temp\{GUID}-78.0.3904.__chrome_updater.exe (21 distinct GUID-prefixed paths)
- **User context recorded by McAfee:** shieldbase\mhill
- **Detecting process:** C:\Windows\System32\svchost.exe
- **Alternative explanation:** A genuine Chrome auto-update is performed by C:\Program Files (x86)\Google\Update\GoogleUpdate.exe, not by a binary named chrome_updater.exe dropped into a per-GUID temp folder. The McAfee GenericRXIT-PW family designation indicates a generic password-stealing variant.
- **Limitation:** All samples deleted by McAfee at detection time; no copy present for static analysis.

Finding F-004 - Offensive PowerShell shellcode-injection script executed on 2018-09-06

- **Confidence:** Confirmed (script block content directly observed in PowerShell EID 4104)
- **Claim Type:** Direct Observation (content) plus Supported Inference (PowerSploit family attribution)
- **Evidence:** Microsoft-Windows-PowerShell/Operational.evtx, EID 4104 (./exports/parsed/evtx/ps_scriptblock_4104.csv filtered via ./analysis/filter_ps2.py)

- **Tool Output:** ./logs/20260604T211038Z_22671_stdout.log
- **User SID:** S-1-5-21-3445421715-2530590580-3149308974-1193 (no local profile)
- **Timestamps (UTC):**
- 2018-09-06 17:13:58.1556017 - script block contains \$s=new-object io.memorystream(,[convert]::frombase64string(H4sIAAAAAAAAA... (H4sIAAA is the magic number for gzip)
- 2018-09-06 17:13:58.5346621 - script block contains set-strictmode -version 2, \$doit = @, function func_get_proc_address { ... \$var_unsafe_native_methods = ([appdomain]::currentdomain.getassemblies()).gettype(microsoft.win32.unsafenativemethods)
- 2018-09-06 17:14:05.3318470 / .4587887 - same script body re-logged
- **Why this matches PowerSploit / Invoke-Shellcode:** The variable name func_get_proc_address, the use of unsafenativemethods reflection to resolve GetProcAddress/VirtualAlloc/CreateThread, and the embedded gzip+base64 shellcode buffer are the canonical scaffold of PowerSploit Invoke-Shellcode.ps1 / Invoke-ReflectivePEInjection.ps1.
- **Alternative explanation:** A sanctioned red-team or penetration test exercise could log identical script blocks.
- **Limitation:** Cannot identify process parent because Sysmon was not yet installed and Security EID 4688 process-create logging is sparse for the relevant time. The base64 buffer was not decoded in this analysis.

Finding F-005 - Suspicious staging directory at C:\ProgramData\staging\

- **Confidence:** Probable
- **Claim Type:** Supported Inference (artifact pattern; not a confirmed malware detection)
- **Evidence:** Directory listing and SHA-256/MD5 of each binary (./logs/20260604T205814Z_13779_stdout.log, ./logs/20260604T205836Z_20110_stdout.log)
- **On-disk component table** (SHA-256 column is truncated to first 16 chars; full values in artifact_manifest.csv):**

FILE PATH	SIZE BYTES	MD5	SHA-256	EVIDENCE SOURCE	NOTES
C:\ProgramData\staging\7za.exe	476672	885e9eb42889ca547f4e351dcde5d3d	b3a70d388488c34d..	EVID-001	7-Zip CLI
C:\ProgramData\staging\NewNotVeriSign.cer	1549	NOT_COMPUTED	47e493034298d628..	EVID-001	Name parodies VeriSign
C:\ProgramData\staging\NotVerisign.cer	1659	NOT_COMPUTED	dc94128e9300c55f...	EVID-001	Name parodies VeriSign

FILE PATH	SIZE BYTES	MD5	SHA-256	EVIDENCE SOURCE	NOTES
C: \ProgramData\staging \lariatca.cer	857	NOT_COM PUTED	a7dc1c864 73b2755...	EVID-001	Lariat CA cert
C: \ProgramData\staging \install_wor mhole\inst all_msadva pi2_32.exe	14183796	85730757c 16a07ce59 6cc5a5237 2b69c	ebb75bbae 3e1298c...	EVID-001	Filename mimics advapi2; dir name install_wor mhole
C: \ProgramData\staging \install_wor mhole\inst all_msadva pi2_64.exe	15184440	49f69b641 b65ba5a4 67ee4436 0a1cfe8	27d496871 6a095a1...	EVID-001	64-bit counterpar t
C: \ProgramData\staging \lrange_lar iat\Lariat-9 .4.1- install.exe	13842063 0	ed080a87c ede8a5ba8 d31b2bb47 4ee9f	70a289edf 326eb7c...	EVID-001	Unknown 138 MB installer
C: \ProgramData\staging \mcollectiv e_agents\ mcollective -puppet- agent-1.13. 0.zip	29667	NOT_COM PUTED	5ce24f6a1f 64d9f2...	EVID-001	Puppet mcollective Puppet- control agent

FILE PATH	SIZE BYTES	MD5	SHA-256	EVIDENCE SOURCE	NOTES
C:\ProgramData\staging\mcollective_agents\mcollective-shell-agent-0.0.2.zip	7768	NOT_COM PUTED	954e8504 3269ec06. ..	EVID-001	Puppet mcollective shell- execution agent (arbitrary shell commands across managed nodes)

- **Why probable rather than confirmed malicious:** Each file individually has a plausible legitimate explanation (administrator tooling, Puppet management, internal CA installs). The collective combination of 7-Zip, advapi2-mimic installers under install_wormhole, an unknown Lariat installer, mcollective shell-agent capable of arbitrary remote command execution, and VeriSign-parody certificates is unusual for a workstation and aligns with attacker tradecraft.
- **Alternative explanation:** A Puppet-managed enterprise environment may legitimately stage tools under C:\ProgramData\staging\.

Finding F-006 - Domain user (SID -1212) added to local Administrators on 2020-02-17

- **Confidence:** Confirmed
- **Claim Type:** Direct Observation
- **Evidence:** Security EID 4732
- **Tool Output:** ./logs/20260604T211148Z_2569_stdout.log
- **Timestamp:** 2020-02-17 08:00:37.4306921 UTC
- **Actor (subject):** shieldbase\BASE-WKSTN-01\$ (the host machine account, NT AUTHORITY\SYSTEM, S-1-5-18)
- **Target group:** BUILTIN\Administrators (S-1-5-32-544)
- **Member added:** S-1-5-21-3445421715-2530590580-3149308974-1212 (domain user; no local profile)
- **Alternative explanation:** Group Policy or System Center push under SYSTEM may legitimately modify local-Administrators membership.

Finding F-007 - Sysmon EDR was deployed on 2019-12-22 by domain admin rsydow-a

- **Confidence:** Confirmed
- **Claim Type:** Direct Observation
- **Evidence:** XML body of \Windows\System32\Tasks\Sysmon Update and on-disk Sysmon64.exe
- **Tool Output:** ./logs/20260604T210403Z_2336_stdout.log
- **Sysmon binary SHA-256:** 80b110b91730729b...
- **Notes:** Sysmon Update task registered by shieldbase\rsydow-a with start boundary 2019-12-22 11:15:00, running SYSTEM, repeats hourly. The Microsoft-Windows-

Sysmon/Operational.evtx channel contains 308,811 events in this image. Sysmon was deployed after the McAfee detection clusters of 2019-05 and 2019-10/11, so it did not observe those deleted samples.

Finding F-008 - Defender real-time protection state events

- **Confidence:** Confirmed (events present) / Probable benign (interpretation)
- **Claim Type:** Supported Inference
- **Evidence:** Microsoft-Windows-Windows Defender/Operational.evtx (./exports/parsed/evtx/defender_events.csv)
- **Tool Output:** ./logs/20260604T210939Z_20080_stdout.log
- **Events observed:** EID 5007 platform install location change (2018-05-04); EID 5010 protection-state changes (2018-05-07, 2018-05-10, 2019-12-17, 2020-09-21); EID 5001 Real-time Protection disabled (2020-09-21 01:01:35)
- **Interpretation:** McAfee VirusScan Enterprise was the active AV; Microsoft Defender normally disables itself when a third-party AV registers with the Security Center.
- **Alternative explanation:** Adversarial Defender disable is not ruled out from EID 5001 alone, but the long-running McAfee install is a stronger explanation in this case.

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Findings Table (Summary)

FINDING	CONFIDENCE	EVIDENCE SOURCE	TOOL OUTPUT PATH	TIMESTAMP UTC	LIMITATION
F-001 Artemis trojans in perfmon-k	Confirmed	McAfee OnAccess ScanLog	./logs/ 20260604 T205909Z _12853_std out.log	2019-05-0 8 / 2019-05-17	AV-only signal
F-002 perfmon-k residual binaries	Probable	sha256sum of on-disk files	./logs/ 20260604 T205838Z _10853_std out.log	n/a	No execution evidence
F-003 84 chrome_up dater trojans (mhill)	Confirmed	McAfee OnAccess ScanLog	./logs/ 20260604 T205933Z _21809_std out.log	2019-10-2 2 to 2019-11-06	Samples deleted by AV
F-004 PowerSploit-style shellcode injection	Confirmed	PowerShell EID 4104	./logs/ 20260604 T211038Z_ 22671_std out.log	2018-09-0 6 17:13:58 - 17:14:05	Process parent not identified
F-005 Staging dir w/ 7za +	Probable	Directory listing +	./logs/ 20260604 T205836Z	n/a	Could be internal tooling

FINDING	CONFIDENCE	EVIDENCE SOURCE	TOOL OUTPUT PATH	TIMESTAMP UTC	LIMITATION
wormhole + Iariat + mcollective		sha256sum	_20110_std out.log		
F-006 Domain SID -1212 added to local Admins	Confirmed	Security EID 4732	./logs/20260604T211148Z_2569_stdout.log	2020-02-17 08:00:37	GPO correlation not performed
F-007 Sysmon deployed by rsydow-a	Confirmed	Task XML + on-disk binary	./logs/20260604T210403Z_2336_stdout.log	2019-12-22	Sysmon installed after detections
F-008 Defender 5001/5010 events	Confirmed (events) / Probable benign (interp)	Defender Operational EVTX	./logs/20260604T210939Z_20080_stdout.log	2018-2020	Consistent with McAfee active

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Claim Support Matrix

FINDING ID	CLAIM	CLAIM TYPE	CONFIDENCE	SUPPORTING EVIDENCE	TOOL OUTPUT PATH	ALTERNATIVE EXPLANATION	LIMITATION
F-001	McAfee detected Trojan family Artemis in 2 files in C:\ProgramData\perfmon-k\ on 2019-05-08 and	Direct Observation	Confirmed	McAfee OnAccessScanLog.txt	./logs/20260604T210509Z_12853_stdout.log	Cloud heuristic false positive	AV log records detection, not execution

FINDING ID	CLAIM	CLAIM TYPE	CONFIDENCE	SUPPORTING EVIDENCE	TOOL OUTPUT PATH	ALTERNATIVE EXPLANATION	LIMITATION
	2019-05-17						
F-002	5 binaries named perfmon-k*.dll/.exe remain on disk in the same directory as the F-001 deletions	Supported Inference	Probable	sha256sum output for the 5 files	./logs/20260604T205838Z_10853_stdout.log	Files could be unrelated legitimate software	Not analyzed against VT or signed-status
F-003	McAfee deleted 84 chrome_updater.exe instances in C:\Users\mhill\AppData\Local\Temp\{GUID}\ between 2019-10-22 and 2019-11-06; threat family Generi	Direct Observation	Confirmed	McAfee log entries with user, path, MD5, threat name	./logs/20260604T205933Z_21809_stdout.log	Genuine Chrome update binary - rejected based on path template under Temp\{GUID} and McAfee family	All 84 samples destroyed by AV

FINDING ID	CLAIM	CLAIM TYPE	CONFIDENCE	SUPPORTING EVIDENCE	TOOL OUTPUT PATH	ALTERNATIVE EXPLANATION	LIMITATION
	cRXIT-PW						
F-004	A Power Shell 4104 script block executed at 2018-09-06 17:13:58 UTC under SID -1193 contains the Power Sploit Invoke-Shellcode scaffold	Direct Observation + Supported Inference	Confirmed	EID 4104 script block text in Microsoft-Windows-PowerShell/Operational.evtx	./logs/20260604T211038Z_22671_stdout.log	Sanctioned red-team exercise	Cannot identify host process; shellcode not decoded
F-005	C:\ProgramData\staging\ contains 7-Zip CLI, two advapi2-mimic installers in install_wormhole, a 138 MB Lariat installer	Direct Observation	Probable	Directory listing and sha256 sum	./logs/20260604T205836Z_20110_stdout.log	Internal Puppet /IT staging	No vendor identification

FINDING ID	CLAIM	CLAIM TYPE	CONFIDENCE	SUPPORTING EVIDENCE	TOOL OUTPUT PATH	ALTERNATIVE EXPLANATION	LIMITATION
	r, mcollective shell-agent, and three VeriSig n-parody cert files						
F-006	Domain SID -1212 added to BUILTI N\Administrators on 2020-02-17 08:00:37 UTC by host machine account (SYSTEM)	Direct Observation	Confirmed	Security EID 4732 row	./logs/ 20260 604T21 1148Z_ 2569_s tdout.log	Legitimate GPO push	No GPO correlation
F-007	Sysmon deployed on 2019-12-22 by shieldbase\rsydow-a	Direct Observation	Confirmed	Task XML body + Sysmon64.exe on disk	./logs/ 20260 604T21 0403Z _2336_ stdout.log	n/a	n/a
F-008	Defender real-time protect	Supported	Confirmed (event) /	Defender Operation	./logs/ 20260 604T21 0939Z_	Adversarial Defender	Single event

FINDING ID	CLAIM	CLAIM TYPE	CONFIDENCE	SUPPORTING EVIDENCE	TOOL OUTPUT PATH	ALTERNATIVE EXPLANATION	LIMITATION
	ion was disabled on 2020-09-21 (EID 5001); disable is most consistent with McAfee being the active AV	Inference	Probable benign (interpretation)	onal EVTX	20080_stdout.log	disable - not ruled out	with no actor

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IOC Table

Note: SHA-256 hashes shown in PDF tables are truncated to first 16 hex chars for visual fit. Full 64-character hashes are in `./analysis/ioc_register.csv` and `./reports/artifact_manifest.csv`. SHA-256 column header below should be read as SHA-256 (truncated).

Full register at `./analysis/ioc_register.csv`.

IOC ID	TYPE	VALUE	SOURCE	FIRST SEEN UTC	CONFIDENCE	NOTES
IOC-001	MD5	1977c126e8e74da4460f41653ca6a6f2	McAfee VSE log	2019-05-08T15:14:04Z	Confirmed	Artemis! 1977C126E8E7; perfmon - kconfigure.exe; AV-deleted
IOC-002	MD5	50cce7d590ca72cfb1f3b5	McAfee VSE log	2019-05-17T22:29:13Z	Confirmed	Artemis! 50CCE7D590CA; perfmon

IOC ID	TYPE	VALUE	SOURCE	FIRST SEEN UTC	CONFIDENCE	NOTES
		522ea5c27b				-k.exe; AV-deleted
IOC-003	MD5	264609c34194d47dbf25d18d4f83d41d	McAfee VSE log	2019-10-22T20:55:41Z	Confirmed	GenericR XIT-PW! 264609C34194; chrome_updater.exe; AV-deleted
IOC-004	MD5	d870ac37e98c1351fd0de3ae404a180d	McAfee VSE log	2019-11-04T19:55:52Z	Confirmed	GenericR XIT-PW! D870AC37E98C; chrome_updater.exe; AV-deleted
IOC-005	MD5	4d17f453b454859f249990be61f36091	McAfee VSE log	2019-11-06T22:01:05Z	Confirmed	GenericR XIT-PW! 4D17F453B454; chrome_updater.exe; AV-deleted
IOC-006	SHA-256	225efff946151ede...	sha256sum on-disk	n/a	Probable	C: \ProgramData\perfmon-k\perfmon-khk.dll
IOC-007	SHA-256	fa76d9059fc4fe10...	sha256sum on-disk	n/a	Probable	C: \ProgramData\perfmon-k\perfmon-ki.dll
IOC-008	SHA-256	fc7f5094bc182a05...	sha256sum on-disk	n/a	Probable	C: \ProgramData\perfmon-k\perfmon-kr.exe

IOC ID	TYPE	VALUE	SOURCE	FIRST SEEN UTC	CONFIDENCE	NOTES
IOC-009	SHA-256	128cdb7c58bfda11...	sha256sum on-disk	n/a	Probable	C:\ProgramData\perfmon-k\perfmon-kvw.exe
IOC-010	SHA-256	8db5e3bdd57ed77f...	sha256sum on-disk	n/a	Probable	C:\ProgramData\perfmon-k\perfmon-kwb.dll
IOC-011	Path	C:\ProgramData\perfmon-k\	McAfee log + filesystem	2019-05-08T15:14:04Z	Confirmed	Drop location for Artemis trojans and 5 residuals
IOC-012	Path	C:\Users\mhill\AppData\Local\Temp\{GUID}-*chrome_updater.exe	McAfee log	2019-10-22T20:55:41Z	Confirmed	21 GUID-prefixed drops
IOC-013	User Context	shieldbase\mhill	McAfee log	2019-10-22T20:55:41Z	Confirmed	User session in which chrome_updater drops occurred
IOC-014	Threat Name	Artemis!1977C126E8E7	McAfee VSE	2019-05-08	Confirmed	family/variant
IOC-015	Threat Name	Artemis!50CCE7D590CA	McAfee VSE	2019-05-17	Confirmed	family/variant
IOC-016						

IOC ID	TYPE	VALUE	SOURCE	FIRST SEEN UTC	CONFIDENCE	NOTES
	Threat Name	GenericR XIT-PW! 264609 C34194	McAfee VSE	2019-10-22	Confirmed	family/variant
IOC-017	Threat Name	GenericR XIT-PW! D870AC 37E98C	McAfee VSE	2019-11-04	Confirmed	family/variant
IOC-018	Threat Name	GenericR XIT-PW! 4D17F45 3B454	McAfee VSE	2019-11-06	Confirmed	family/variant

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Gaps and Limitations (Unknowns)

- Unknown - original delivery vector of the perfmon-k* files in May 2019. Sysmon was not yet installed; Security 4688 process-create logs are sparse for that period.
- Unknown - identity of domain user behind SID -1193 who ran the 2018-09-06 PowerShell. The SID is not in this host local ProfileList or SAM. Account name only resolvable against the domain controller (outside this evidence boundary).
- Unknown - identity of SID -1212 added to local Administrators on 2020-02-17. Same constraint.
- Unknown - whether Lariat-9.4.1-install.exe (138 MB) is malicious. Not scanned in this analysis; no signed-status verification.
- Cannot be determined from available evidence - whether any outbound network transfer of credentials or data occurred. SRUM extracted but not parsed (SrumECmd not installed); no network capture in scope.
- Cannot be determined from available evidence - whether the mcollective-shell-agent zip was ever extracted and executed.
- Evidence that would resolve the above:
- Decode the gzip+base64 buffer logged in EID 4104 at 2018-09-06 17:13:58 (offline shellcode analysis).
- Submit perfmon-k*.dll/.exe SHA-256 hashes to VirusTotal or a multi-AV scanner.
- Obtain SRUDB.dat parse output (per-app network bytes) on a host where SrumECmd is available.
- Query the domain DC to resolve SIDs -1193 and -1212.
- Compare residual C:\ProgramData\perfmon-k\ files against a clean shieldbase host baseline.

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Reproducibility Appendix

- **Command Ledger Location:** ./logs/command_ledger.csv

- **Total commands logged:** 81
 - **Failed commands:** 19 (each failure has a documented cause: mmls returned non-zero because EVID-001 is a logical NTFS volume with no MBR partition table; PECmd.dll and SrumECmd.dll are not installed on this SIFT instance; yara binary is not in PATH for sudo; one head-induced SIGPIPE was non-fatal)
 - **Rerun / retry commands noted:** 0
 - **Unresolved command failures:** 0
 - **Statement:** The command ledger audit passed.
 - **Audit output:** ./logs/20260604T211230Z_3496_stdout.log
 - **Output directories used:** ./analysis/, ./exports/, ./exports/parsed/, ./logs/, ./reports/
 - **Evidence integrity:** ewfverify stored MD5 matched recomputed MD5 (./logs/20260604T205418Z_15768_stdout.log); EVID-001 mounted strictly read-only for the entire analysis; all forensic commands executed through /home/sansforensics/claude/scripts/run_logged.sh per the SIFT protocol.
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Confidence Level Definitions

CONFIDENCE	DEFINITION
Confirmed	Directly supported by raw or parsed forensic artifact.
Corroborated	Supported by two or more independent artifacts.
Probable	Supported by one artifact plus strong contextual evidence.
Hypothesis	Plausible but not yet proven; must include the next validation step.
Rejected	Tested and unsupported.

This report was produced as part of an active digital forensics investigation. All findings are based on evidence present at the time of analysis. Evidence integrity maintained per chain-of-custody protocol — source images not modified.